

Mradul Singh

AI / ML Engineer · LLM Agents, RAG & Distributed Systems

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Summary

- Software engineer with **~3 years** of experience shipping production **AI/agent systems** and scalable distributed backends.
- Built an **in-product LLM assistant**, an internal **multi-agent coding & PR-review agent**, and a **Temporal-based agent-evaluation harness**.
- Experienced across the modern LLM stack including **Claude, Bedrock, LangGraph, RAG, LLM-as-judge evaluations, and OpenTelemetry**.
- Strong backend engineering foundation in **Java/Python microservices, Kafka event pipelines, authorization models**, and production-scale distributed systems.

Experience

- **Software Engineer, Accolade** (Jan'25- Present)
 - Built “**Athena**,” an in-product AI assistant embedded in a Java 21 backend on **Anthropic’s Claude Managed Agent API** — inbound messages over **Aply AMQP**, dispatched on virtual-thread executors, streamed to the web client via the **Vercel AI Data-Stream protocol**; designed an in-process **@AgentTool** framework (Spring AI) exposing 7+ domain tools and reusable agent “skills.”
 - Architected **iris**, a Python agent backend: a **LangGraph** supervisor coordinating specialized sub-agents on **AWS Bedrock**, with **RAG** over a gRPC knowledge base and **LangSmith** prompt-versioning/tracing; added session-aware **OpenTelemetry** spans propagated (W3C tracecontext) from Python into the Java services.
 - Built “**Strike**,” an internal engineering agent similar to **Claude Tag** (Slack + FastAPI/SSE, multi-agent runtime, PostgreSQL full-text memory) featuring an **LLM-as-judge evaluation suite**, plan-mode auto-review, and per-iteration token/cost tracking; shipped its **AI PR-review** product (deep reviews with feedback learning, AI-generated PR descriptions) — adopted across **18 repositories**.
 - Founded **hera**, a gRPC + **Temporal AI-agent evaluation & simulation harness**: generates synthetic property-management datasets with **NVIDIA NeMo** and runs durable, replay-safe eval workflows (batch + streaming) that simulate full technician and prospect-conversation lifecycles against live services.
 - Backed the AI features with production distributed-systems plumbing — a **Kafka transactional outbox** for exactly-once event publishing and **multi-tenant, replay-safe Temporal context propagation** — across a Java 21 modular monolith and Python services.
 - Shipped end-to-end **CQRS** product domains in the Java monolith — property inspections (gRPC lifecycle, immutable versioned templates, Temporal-driven expiration & report generation), work orders, and technician timesheets — plus a **pgvector**-backed knowledge-base module (extracted into standalone Maven modules behind a clean API) for semantic retrieval.
 - Owned supporting platform work end-to-end: event-driven **Kafka** domain pipelines, AWS infrastructure-as-code (ECS/Fargate, IAM, S3→SQS, Lambda via **Terraform/CDK**), and **Databricks** data flows (Lakebase, DLT, Unity Catalog).
 - Rolled out reusable **in-house AI PR-review/description GitHub Actions** org-wide; contributed 2,500+ commits / 450+ merged PRs across 15 repositories in ~16 months.
- **Software Developer, PeakFlo** (Jan'24- Jan'25)
 - Built and optimized core **payment** processing on a TypeScript platform (Firebase, GCP), cutting internal payment-API latency around **50 ms** and revamping a business-critical payment-gateway integration.
 - Led development of a new **travel-&-expense** module — data model, APIs, and user-facing flows — expanding the platform into a new product segment.
 - Integrated **accounting/ERP** systems (QuickBooks/Xero-style sync) to auto-post expenses and payments and cut manual bookkeeping.
 - Built configurable **approval workflows** for expense and payment authorization (multi-level approvers, spend limits, policy checks).
- **Backend Engineer, Ostello AI** (Sep'23- Jan'24)
 - Scaled backend services from **1K to 100K users** by fronting them with an **nginx** reverse proxy using IP-hash load balancing (Node.js/NestJS, GraphQL, PostgreSQL).
 - Cut database connection setup/teardown from **10 ms to 3 ms** by introducing **connection pooling** (pg-pool), improving backend throughput under load.
 - Built a secure authentication system with **multi-factor authentication (MFA)** and email verification.
 - Designed RESTful & **GraphQL** APIs and a MongoDB-backed **event-logging pipeline** (~10K events/hour) for request tracing and analytics.

Key Projects

- **agentctl** — Durable, observable runtime for LLM agents *Python, Postgres, S3, OTel*
 - Crash-resumable execution via a **write-ahead log** + deterministic state machine — interrupted runs resume from the last committed step with **exactly-once, idempotency-keyed tool side-effects**.
 - Durable **human-in-the-loop**: suspend an agent run for hours awaiting approval, then resume it safely without losing context.
 - **OpenTelemetry**-native tracing + **per-tenant token-cost attribution** + **CI eval-gates** that fail the build on quality regressions.
 - Provider-agnostic adapter (Claude / Bedrock / OpenAI) wrapping any agent loop; fault-injection harness that kills the process mid-tool-call.
- **fga-gateway** — ReBAC auth proxy for LLM agents & MCP *Go, OpenFGA, MCP*
 - Enforces per-object, **relationship-based (Zanzibar/OpenFGA)** permissions on every agent tool call and MCP request. An agent can access only what the end user is authorized to.
 - **Policy-as-code** DSL mapping tool-call intents to authorization checks, with safe delegation and a full allow/deny **audit trail**.
 - Deployed as a **sidecar proxy** with sub-millisecond check caching and **fail-closed** defaults; treats retrieved content as untrusted so prompt injection can't bypass authz.
 - **Threat-model design doc** on the agent-authorization boundary.
- **logcask** — Raft-replicated distributed commit log *Java, Raft*
 - Implemented an append-only **segmented-log storage engine** — memory-mapped segments, batched **fsync**, a sparse offset index, and background compaction — exposing an ordered, durable record stream.
 - Wrote a **Raft** consensus layer from scratch: leader election, log replication, snapshotting/log compaction, and dynamic membership changes for strongly-consistent replication across a cluster.
 - Proved correctness with a **Jepsen-style linearizability harness** that injects network partitions and kills the leader mid-write, asserting zero lost or duplicated records across failovers.
 - Built a producer/consumer API with **consumer groups**, offset commits, and **exactly-once** delivery via idempotent producers; added backpressure and configurable durability (acks).
 - Instrumented with **OpenTelemetry** and load-tested sustained throughput/latency.
 - Wrote a **design doc** on the consistency model, replication protocol, and failure modes.

Technical Skills

- **General:** Distributed Systems, Event-Driven Architecture, Microservices, CQRS, System Design, Agentic AI Infrastructure, RAG Systems, Multi-tenancy, ReBAC Authorization, Optimization Systems, Scalable Databases
- **AI / LLM:** Anthropic Claude (Agent SDK, Claude Managed Agent), AWS Bedrock (Converse, Nova, Guardrails), LangGraph, LangChain, LangSmith, RAG & Hybrid Retrieval, LLM-as-judge Evals, Agent Orchestration, MCP, pgvector, NVIDIA NeMo, Prompt & Context Engineering
- **Programming Languages:** Python, Java 21, TypeScript, JavaScript, SQL, Protobuf, C, C++, HTML/CSS
- **Backend & Infrastructure:** Spring Boot, Spring AI, FastAPI, Node.js, Express.js, NestJS, gRPC/Protobuf (Buf), REST & GraphQL API Design, Apache Kafka, Temporal, Timefold Optimization, Docker, Kubernetes, Nginx, Terraform/CDK, CI/CD (GitHub Actions), Reverse Proxying
- **Frontend & Tools:** React.js, Redux Toolkit, TanStack Query, Tailwind CSS, Vite, Figma, Git, GitHub
- **Databases, Cloud & Observability:** PostgreSQL (+ pgvector; schema optimization & tuning), MongoDB, MySQL, Redis/Valkey, Databricks (Lakebase, DLT, Unity Catalog), AWS (ECS/Fargate, Lambda, S3, SQS, Elastic Beanstalk), GCP, Firebase, OpenFGA/Auth0 FGA, OpenTelemetry, Datadog, Grafana/Loki

Education

- B.Tech + M.Tech (Dual Degree) - Indian Institute of Technology (BHU), Varanasi 2021 – 2026

Positions of Responsibility

- **Tech Head** — Technex & Spardha, IIT (BHU) Varanasi 2023 – 2024
 - led technology for the institute's flagship technical and sports festivals.
- **Core Member** — Club of Programmers (COPS), IIT (BHU)

Achievements

- **Competitive Programming:** CodeChef 4-star (max 1897) · Codeforces Pupil (max 1510) · rank 514/30K, Codeforces Round 860 · rank 44/20K, CodeChef Starters 80.
- **Scholastic:** Top 5%ile JEE Main · AIR 2K, AKTU (of 0.12M) · KVPY & PRMO qualifier · Team Rank 3, Flipkart GRiD 4.0 (100+ teams).